CODEVEDA - HACKATHON 2026

- **Problem Statement Title: AI-Powered Personalized Learning Platform For Government Schools**
- **Theme: SDG 4 – Quality Education**
- **Team Name: Code Crew**
- **Team Leader’s Name: Keerthana K**
- **Team Members Name : Anisha Sherin Y, Naveen Kumar Reddy, Sandesh M K, Akash T**

IDEA TITLE : AI-Powered Personalized Learning Assistant

Proposed Solution :

Develop a web platform that uses AI and learning analytics to personalize education for students in government schools.

How it Works:

- Students take quizzes and lessons
- AI analyzes performance and learning patterns
- System identifies weak topics
- Personalized study plans are generated
- Teachers monitor progress via dashboard

Innovation:

- AI-based learning gap detection
- Adaptive quizzes
- Personalized content recommendation
- Data-driven teacher insights

TECHNOLOGY & METHODOLOGY

Technologies Used:

- Frontend
  - React.js
  - HTML, CSS, JavaScript
- Backend
  - Node.js + Express, JWT
- Database
  - Azure SQL
- AI Module
  - Azure Open AI
  - Azure AI Studio
  - Azure App Service
- Cloud Platform
  - Microsoft Azure

Azure Services:

- Azure App Service → Backend hosting
- Azure Static Web Apps → Frontend hosting
- Azure SQL Database → Student data storage
- Azure Machine Learning → AI models
- Azure AI Services → future AI tutor chatbot

System Architecture:

Auth (JWT Login)
↓
Students (Profile data)
↓
Quiz(Submit Scores)
↓
AI(Study plans)
↓
Teacher(Analysis and alerts)
↓
Generation of Study Plan

FEASIBILITY AND VIABILITY

Technical Feasibility:

- Built on scalable cloud infrastructure
- Uses widely adopted technologies
- Easy integration with existing school systems
- Web-based system accessible on any device

Economic Feasibility:

- Cloud infrastructure reduces hardware costs
- Open-source tools reduce development cost
- Scalable solution for thousands of students

Potential Challenges:

- Limited internet connectivity in rural areas
- Data privacy and security concerns
- Teacher adaptation to new technology

Operational Feasibility:

- Simple interface for students and teachers
- Requires minimal training
- Accessible through browser or mobile device

Risk Mitigation Strategies:

- Offline learning support
- Secure data encryption
- Pilot deployment in selected schools
- Continuous monitoring & feedback

Long-Term Sustainability:

- Scalable cloud infrastructure
- Continuous AI model improvement
- Government & institutional partnerships

IMPACT AND BENEFITS

Potential Impact on Target Audience

Students:

• Personalized learning and better understanding of weak topics

• Improved engagement and self-paced learning

Teachers:

• Real-time insights into student performance

• Easier identification of struggling students

Education System:

• Scalable solution for government schools

• Improved learning outcomes through data-driven insights

Benefits of the Solution

Social:

• Reduces learning gaps among students

• Promotes equal access to quality education

• Encourages self-paced and inclusive learning

Economic:

• Cost-effective digital learning platform

• Reduces need for additional tutoring resources

• Scalable solution for large classrooms and schools

Environmental:

• Reduces paper usage through digital quizzes and materials

• Supports eco-friendly and sustainable learning practices

• Minimizes physical resource consumption in education systems

RESEARCH AND REFERENCES

Research Sources

• Studies on AI-based personalized learning systems

• Research on educational data analytics and adaptive learning

Technologies Referenced

• React

• Node.js

• Microsoft Azure

AI / ML Libraries

•open AI

•Azure AI studio

Additional Resources

• Educational datasets from Azure SQL

• Microsoft documentation and AI learning resources